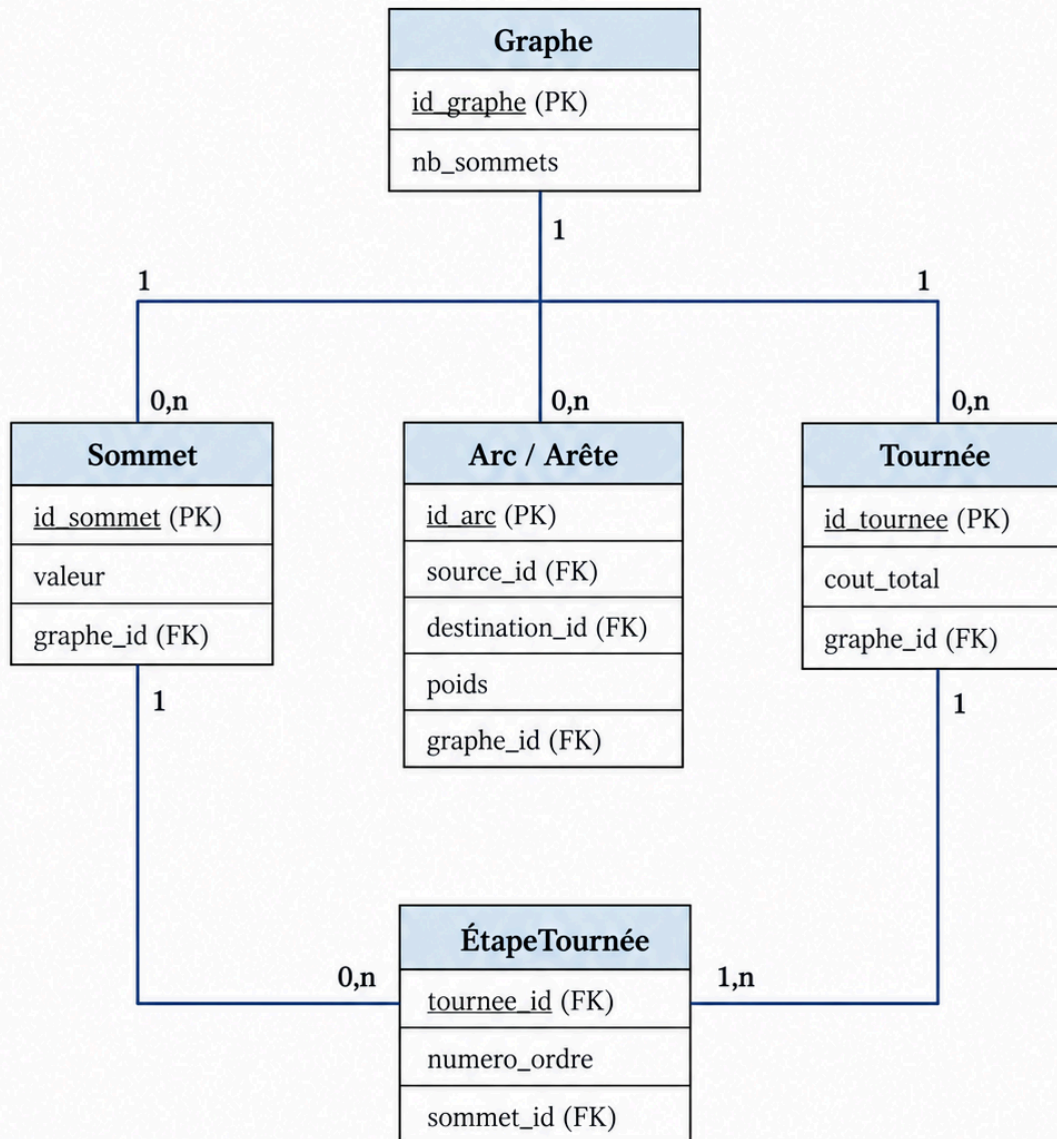


Base de données – Diagramme E/A



Légende

1 : un et un seul
0,n : zéro, un ou plusieurs

(PK) : clé primaire
(FK) : clé étrangère

Implementation en SQL:

-- init_db.sql

-- Script d'initialisation de la base de données TourneeFutee

CREATE DATABASE IF NOT EXISTS tourneefutee;

USE tourneefutee;

-- Table Graphe

```
CREATE TABLE IF NOT EXISTS Graphe (  
    id_graphe INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    nb_sommets INT UNSIGNED NOT NULL DEFAULT 0,  
    est_oriente TINYINT(1) NOT NULL DEFAULT 0  
);
```

-- Table Sommet

```
CREATE TABLE IF NOT EXISTS Sommet (  
    id_sommet INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    valeur VARCHAR(100) NOT NULL,  
    graphe_id INT UNSIGNED NOT NULL,  
    FOREIGN KEY (graphe_id) REFERENCES Graphe(id_graphe) ON DELETE CASCADE  
);
```

-- Table Arc

```
CREATE TABLE IF NOT EXISTS Arc (  
    id_arc INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    source_id INT UNSIGNED NOT NULL,  
    destination_id INT UNSIGNED NOT NULL,  
    poids FLOAT NOT NULL DEFAULT 0,  
    graphe_id INT UNSIGNED NOT NULL,  
    FOREIGN KEY (source_id) REFERENCES Sommet(id_sommet) ON DELETE  
CASCADE,  
    FOREIGN KEY (destination_id) REFERENCES Sommet(id_sommet) ON DELETE  
CASCADE,  
    FOREIGN KEY (graphe_id) REFERENCES Graphe(id_graphe) ON DELETE CASCADE  
);
```

-- Table Tournee

```
CREATE TABLE IF NOT EXISTS Tournee (  
    id_tournee INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    cout_total FLOAT NOT NULL DEFAULT 0,  
    graphe_id INT UNSIGNED NOT NULL,  
    FOREIGN KEY (graphe_id) REFERENCES Graphe(id_graphe) ON DELETE CASCADE  
);
```

```
-- Table EtapeTournée
CREATE TABLE IF NOT EXISTS EtapeTournée (
  tournée_id INT UNSIGNED NOT NULL,
  numero_ordre INT UNSIGNED NOT NULL,
  sommet_id INT UNSIGNED NOT NULL,
  PRIMARY KEY (tournée_id, numero_ordre),
  FOREIGN KEY (tournée_id) REFERENCES Tournee(id_tournée) ON DELETE CASCADE,
  FOREIGN KEY (sommet_id) REFERENCES Sommet(id_sommet) ON DELETE CASCADE
);
```